- Abstract:
    - Refers to “the standard objective” which could mean multiple things, suggest e.g. “standard objective of maximising expected return”.
- Ch1:
    - P2: Explain clearly the multi-objective setting, including what is known when, and how this interacts with on-line planning with MCTS methods. Consider using the unknown weights scenario instead of decision support as motivation.
    - P3: either here or elsewhere justify the choice of linear utility function and discuss whether any of the work could be applied to decision support scenarios where the utilites are non-linear or better expressed as preference functions
    -
- Ch2:
    - P12: Explain which of these bounds is better and why. Give some intuition about simple regret bounds, what constitutes a good bound, etc.
    - P13: “States are sampled according to a transition distribution that depends **only** on the current state and current action being taken (the Markov assumption).”
    - P15: In a finite-horizon MDP, the policy needs to condition on the timestep (as do the value functions defined later).
    - P17: It’s confusing to include RL here. You’re just doing sample-based on-line planning.
    - P18: “The optimal policy is the policy that maximises the objective function J, and can be shown to be deterministic.” There can be multiple optimal policies, some of which may be stochastic though there always exists at least one deterministic optimal policy.
    - P39: discuss implications of assuming linear scalarisation
    - P41: the MCCS or a MCCS?
- Ch3:
    - “replacing the Shannon entropy regulariser **with** other entropy regularisers”
    - P53: Reiterate here the motivation for online MO planning.
- Ch4:
    - P56: “In maximum entropy reinforcement learning (Section 2.3.1), entropy is added
      directly to the objective function rather than used as a constraint to remove ambiguity between multiple optimal policies, as in inverse reinforcement learning.” What’s the difference if the constraint will be turned into a penalty with a Lagrangian?
    - P59: Again, isn’t UCT’s “exploitation” in effect good local exploration?
    - P62: which of these are not satisfied by UCT or MENTS? Why exploit if we care about simple regret?
    - P63: “Uses a Boltzmann search policy with decayed temperature similar to [8], but includes an additional uniform exploration term similar to MENTS” Why? Very procedural with no motivation, which of the above problems are you solving? Why do we need two exploration terms?
- Ch5:

    - Why do we need this approach based on ball? Why not learn separate values for each objective and dot-product them with the weights?
    - Again, be clear about the motivation here, and consider switching to unknown weights scenario.
    - P132: “It is worth noting that in multi-objective MCTS, the quality of the convex hull at the
    - root node is the primary concern.” Why? This is a big unsupported claim
    - P132: “A more efficient online approach is therefore to run the multi-objective tree search from the initial state, and subsequently follow a single-objective MCTS algorithm in the scalarised MDP with reward $R(s,a; w) = w^{\top}R(s,a)$, where w is inferred from the choice made at the root node.” The choice may not disambiguate w, which is another reason why unknown weights would give a cleaner motivation. Also, explain why building an MCTS tree is a good way to use the offline planning budget, compared to just learning better prior policies and value functions that apply in any state. Finally, if this only makes sense assuming a deterministic initial state, that needs to be stated clearly up front.
    - P134: Why is improvement not monotonic? (later plots as well)
- Ch6:
    - P148: concave -> convex
    - P163: correct x-axis on figure size to env size/width, not search time. Also ensure Fig 6.10 is discussed in the text, and the non-monotic trends and fully explained. (also applies to Fig 5.7 on page 138)

- Appendix D:
    - P208 -Fig D.1 - is this really a CDF? Not in the conventional sense? Suggest to explain more or reword

Multiple pages: Fix broken cross-refs - a search for ‘??’ found 10 of them.

Multiple places: BayesOpt:

BayesOpt is one of many possible parameter / hyper-parameter packages - say a bit more about why this was chosen (convenience of C++ implementation etc) and whether the parameter tuning generally made much of a difference. Also consider whether these are parameters or hyper-parameters (arguably these are just parameters).